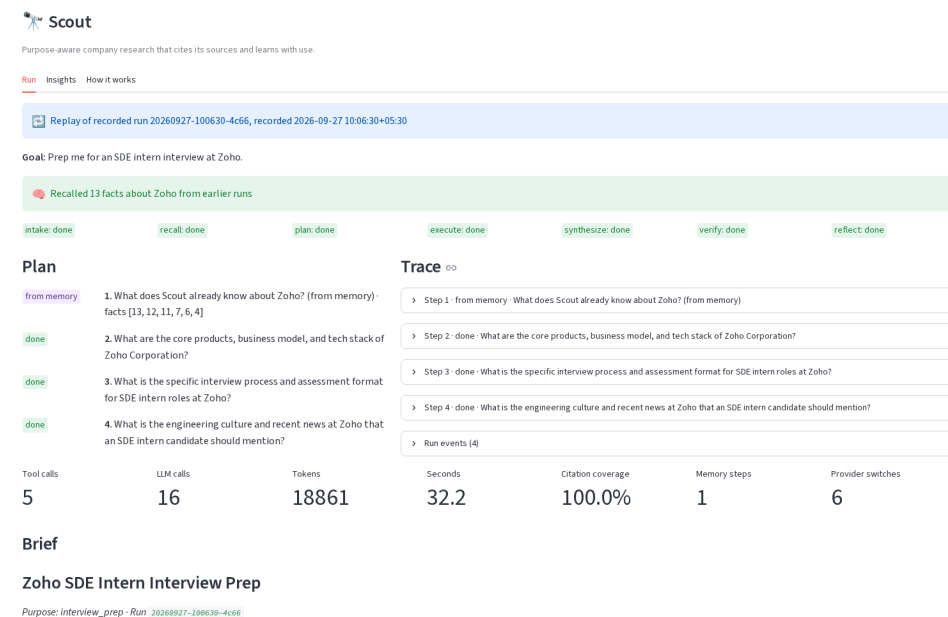
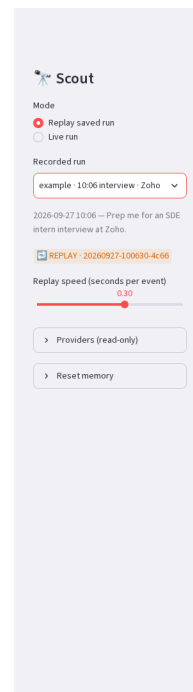




Research that depends on *why* you are asking

- Same company, different purpose
→ different facts: a **sales rep**, a **competitor analyst** and an **interview candidate** need different briefs about Zoho
- Most agents run one generic prompt. Scout turns the purpose into a **playbook-guided plan**, cites every claim, checks its own work, and **learns with use**



Architecture

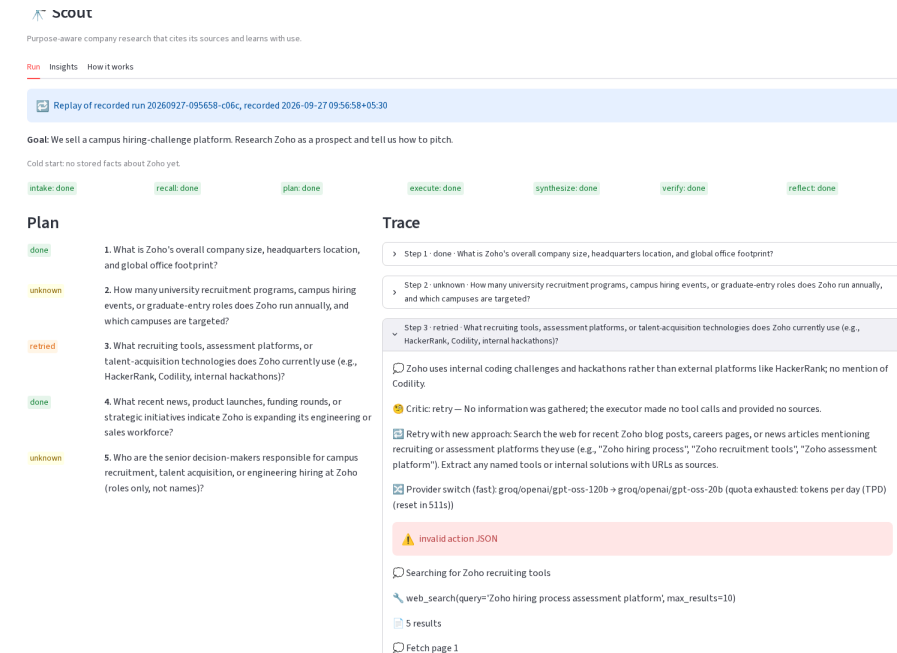
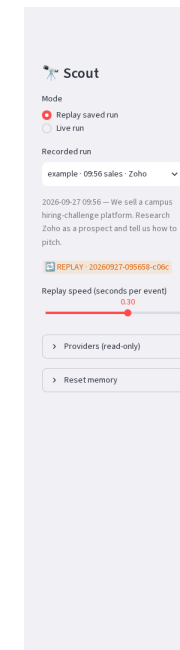
Hand-written state machine (no framework). The orchestrator is a generator of events → CLI, UI and trace all consume the same stream.

Intake → Recall → Plan → [Execute ⇌ Critique]* → Synthesize → Verify → Reflect

Layer	Choice
LLM	OpenAI-compatible wrapper, JSON + Pydantic + 1 repair; router across 3 Groq models + Gemini (free tiers)
Tools	Tavily → DuckDuckGo search, SSRF-guarded fetch (httpx + trafilatura), <code>recall_memory</code>
Memory	SQLite: facts, source scores, lessons, feedback, runs
UI	Typer + Rich CLI, Streamlit with live runs and replay

Agentic patterns in action

- **Plan-and-Execute**: purpose-shaped plan from playbook + lessons
- **ReAct** per step: think → search / fetch / recall → observe → cited findings
- **Critic routing**: complete · **retry** (new approach) · **follow-up** (replan) · unknown
- **Verifier**: citations must exist; **numbers must appear in a cited snippet**; 1 revision, then Unknowns
- **Reflexion**: lessons written after each run, voted on later



Learns with use

1. **Fact memory**: verified facts with source + freshness (7 d stable, 2 d news)
answer steps for free
2. **Source reliability**: critic ratings + 👍/👎 → $(\text{useful}+1)/(\text{useful}+\text{useless}+2)$ re-ranks search
3. **Strategy lessons**: top 3 per purpose injected into the next plan

Same goal ("Prep me for an SDE intern interview at Zoho"), before → after memory:

	tool calls	LLM calls	tokens	seconds
no memory	11	27	38,190	122
1 step from memory, 6 facts	5	16	18,861	32

Results and next steps

- **Evals** (9 recorded runs with the verifier): 100% citation coverage; 13 claims flagged, 5 moved to Unknowns instead of published
- **Safety**: untrusted-content wrapping, SSRF guard, roles only (no profiles or contact data), budgets that end in honest partial briefs, traced provider failover
- **Quality**: 238 offline tests; replayable runs; zero-key example replays

Next: scheduled competitor tracking · vector memory · parallel steps · Scout as an MCP server

Repo: `scout-agent` (README → quickstart in 5 commands) · MIT